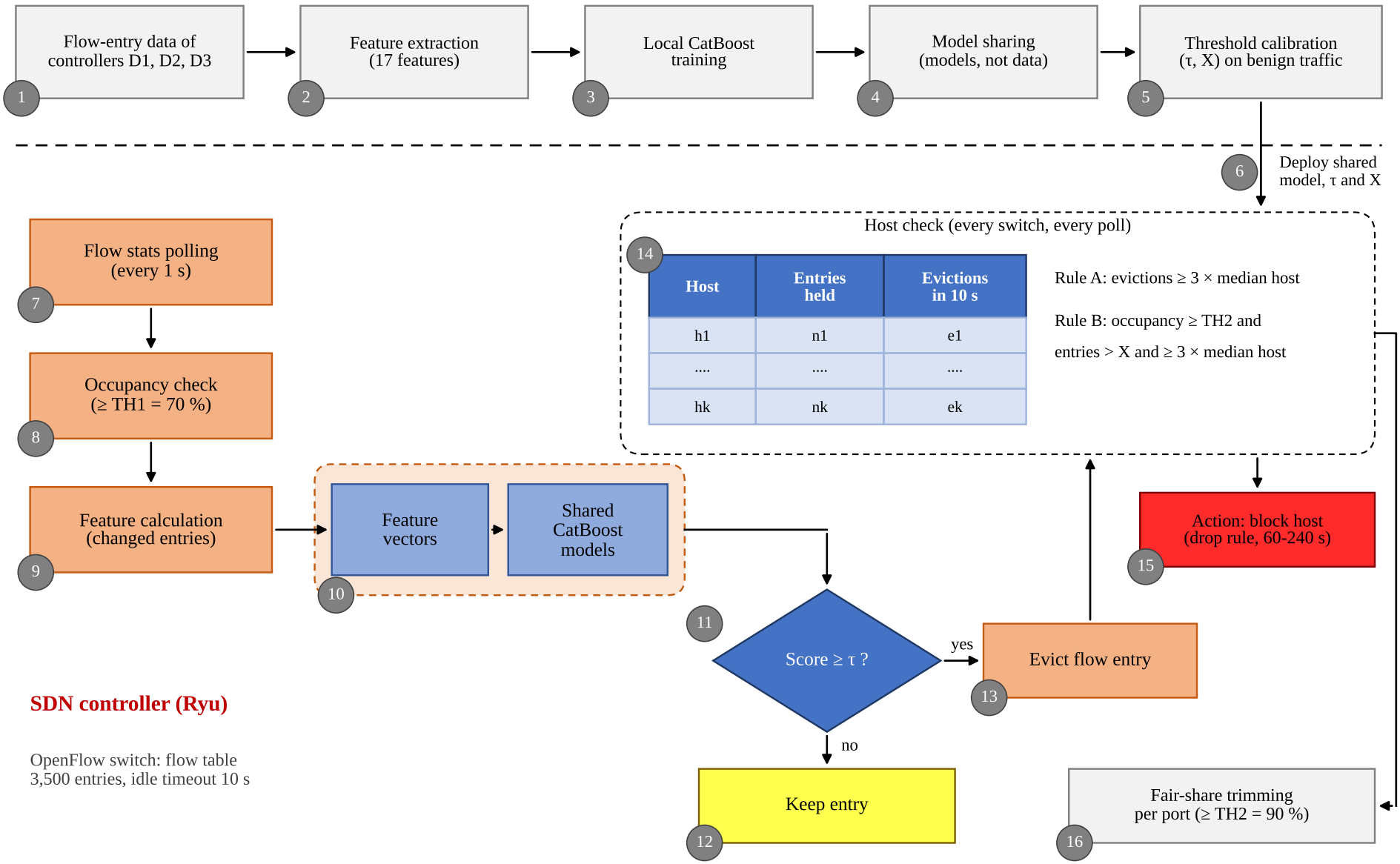
1
Flow-entry data of controllers D1, D2, D3
2
Feature extraction (17 features)
3
Local CatBoost training
4
Model sharing (models, not data)
5
Threshold calibration (τ, X) on benign traffic
6
Deploy shared model, τ and X
7
Flow stats polling (every 1 s)
8
Occupancy check (≥ TH1 = 70 %)
9
Feature calculation (changed entries)
10
Feature vectors
Shared CatBoost models
11
Score ≥ τ ?
yes
no
12
Keep entry
13
Evict flow entry
14
Host check (every switch, every poll)
Host
Entries held
Evictions in 10 s
h1
n1
e1
....
....
....
hk
nk
ek
Rule A: evictions ≥ 3 × median host
Rule B: occupancy ≥ TH2 and entries > X and ≥ 3 × median host
15
Action: block host (drop rule, 60-240 s)
16
Fair-share trimming per port (≥ TH2 = 90 %)
SDN controller (Ryu)
OpenFlow switch: flow table 3,500 entries, idle timeout 10 s